

Carles Antón Blasco

Technical Artist

Audiovisual Engineer and Programmer with extensive experience in pipeline tool development, real-time interactive systems, and graphics programming. Proven track record of bridging the gap between creative teams and technical implementations using Python, C++, and Unreal Engine. Experienced in building performant shaders, optimizing creative workflows, and managing complex digital art pipelines.

Work Experience

Machine Learning Engineer @ Upshelf (Dec 2019 - Dec 2025)

- > Architected high-performance Python pipelines and scalable AWS infrastructure, significantly optimizing processing efficiency and workflow speed.
- > Engineered modular software architectures to solve complex entity-matching challenges across digital and physical data environments.
- > Developed maintainable, modular codebases and comprehensive technical documentation to enforce best practices and streamline cross-team development.

WARE Reinforced @ Matriu.id (May 2022)

- > Developed real-time image processing and data pipelines to dynamically drive 3D visuals and generative content for live performances.
- > Engineered custom interactive tools, sensor-driven controllers, and QR response systems using modular code architecture.
- > Acted as the core technical-art bridge, translating creative concepts into performant real-time systems.

O.M.I. (One More Intelligence) @ Irene Pérez Muñoz (Feb 2023)

- > Integrated Kinect 2.0 depth-tracking data into real-time 3D environments to map live performer movement directly into scenographic visuals.
- > Optimized real-time visual performance and tracking pipelines to ensure zero-latency interaction during live shows.

WARE PROTOTYPE @ Matriu.id (Jun 2023)

- > Engineered real-time interactive systems using body sensors and Kinect 2.0 depth-tracking to process live physical input into 3D world dynamics.
- > Designed and implemented custom real-time shaders and dynamic materials to procedurally react to performer depth data and lighting conditions.
- > Optimized spatial tracking pipelines and interactive graphics to ensure smooth performance during high-traffic live showcase environments.

Contact

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Technical Skills

Programming & Scripting

Python	●●●●●
C++	●●●○○
GLSL	●●●○○
JavaScript	●●●○○
SQL	●●●○○

Real-Time & Engine Tools

Unreal Engine	●●●○○
TouchDesigner	●●●○○
PureData/Max	●●○○○

Math & Systems

Git	●●●●○
Arduino	●●●●○
Linear Algebra	●●●●○
Vector Math	●●●●○
AWS Cloud	●●●○○

Languages

- Catalan (Native)
- Spanish (Native)
- English (B2)

Sorolla, Una Nova Dimensió @ Layers of Reality (Jun 2023)

- > Built a multiplatform real-time application and custom generative imaging pipeline to process live user inputs into styled visual assets.
- > Engineered an end-to-end automated pipeline integrating image capture, AI style transfer, and QR-based distribution for high-throughput public use.

The Great Library of Tomorrow @ Layers of Reality (Nov 2023)

- > Served as the primary technical bridge between creative directors, artists, and software engineers to translate complex artistic concepts into functional real-time tools.
- > Engineered an automated real-time imaging pipeline and multiplatform application to seamlessly ingest, process, and combine user assets.

De Montmartre a Montparnasse. Artistes catalans a París, 1889-1914 @ Picasso Museum (Nov 2024)

- > Architected and implemented an interactive exhibition application to sync real-time audio and visual assets into a cohesive media experience.
- > Collaborated closely with museum curators and visual designers to deliver an intuitive, high-reliability interface for diverse public audiences.

Espigolar per transformar - Espigolar per transformar @ Fundació Espigoladors (Jun 2025 - Set 2025)

- > Designed and built custom generative visual tools in TouchDesigner to produce a high-end audiovisual piece.
- > Worked directly with the Creative Director to translate artistic vision into real-time rendered assets.
- > Engineered the complete audiovisual production and sound design pipeline, optimizing real-time output for an immersive exhibition setting.

Espigolar per transformar - El voluntariat com a resposta @ Fundació Espigoladors (Jun 2025 - Set 2025)

- > Prototyped, designed, and assembled custom PCB hardware and 7-segment display arrays to drive real-time data visualization.
- > Integrated physical electronics with real-time data feeds, managing full hardware-software integration for a high-visibility public installation.

Creative Technologist & Shader Developer @ Independent Projects (Oct 2025 - Jul 2026)

- > Designed and coded custom GLSL shaders to generate complex, performant procedural artwork and real-time visual effects.
- > Built custom web interfaces and interactive front-ends to provide intuitive controls for real-time graphics and user interaction.
- > Applied core mathematical concepts (trigonometry, linear algebra, vector math) to optimize shader performance and rendering efficiency.

Education

ESO (2009-2013)
IES Puig De La Creu,
Castellar del Vallès

Baccalaureate (2013-2015)
IES Puig de la Creu, Castellar
del Vallès

**University Degree -
Audiovisual Systems
Engineering** (2015-2019)
Universitat Pompeu Fabra,
Barcelona

**Real-time Shaders - Code
Art Together** (2020)
CODAME

**3D Environment creation for
Virtual Production using
Unreal Engine** (2025)
CIFO L'Hospitalet de
Llobregat